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Seamless light system for thresholds, cabinets, and baseboards

Abstract

A lighting system integrated into a door and adjacent baseboard which provides a seamless light between the door and baseboard. Multiple lighted strips integrated into channels within the door and baseboard provide a concealed lighting system. By utilizing a concealed hinge, retractable conduit, and light spacing, the lighting system provides a consistent illumination.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to light fixtures. More specifically, the present invention is a lighting fixture for providing consistent illumination beneath doors, baseboards, and cabinets.

BACKGROUND OF THE INVENTION

(2) In modern interior design, lighting plays a crucial role not only in providing illumination but also in enhancing the aesthetic appeal of spaces. As architectural designs evolve, there is a growing demand for innovative lighting solutions that integrate seamlessly with interior elements, such as doors and baseboards, to create cohesive and visually appealing environments. Traditional lighting systems often focus on illuminating specific areas of a room, such as ceilings, walls, or floors. However, the junction between doors and baseboards remains an underutilized space for lighting applications.

(3) The prior art typically involves lighting systems that illuminate specific sections of a room, but these systems often fail to address the junction between the bottom edge of a door and the adjacent baseboard. As a result, gaps in illumination can occur, disrupting the visual continuity and flow of lighting across the room. Some solutions in the prior art integrate lighting into baseboards or door frames, but these often lack the ability to provide continuous illumination across the gap. Additionally, such systems frequently require multiple power sources, which increases the complexity of installation and maintenance.

(4) The prior art fails to provide a mechanism that allows for seamless lighting across the door-baseboard gap, which leads to interruptions in lighting and detracts from the aesthetic value of the space. In environments that prioritize design continuity and minimalism, the inability of the prior art to address these issues is particularly problematic.

(5) The present invention addresses the shortcomings of the prior art by providing a lighting system that delivers seamless, continuous illumination between the bottom edge of a door and the adjacent baseboard. The present invention features a baseboard with a protrusion, wherein a light strip is positioned on the bottom-facing surface of the protrusion. This configuration allows the light to illuminate the space directly above the floor, creating a smooth and uninterrupted lighting effect.

(6) The present invention further includes a concealed hinge and a retractable conduit that links the light strip on the baseboard to a corresponding light strip on the bottom edge of the door. The

retractable conduit ensures that the lighting connection is maintained as the door moves, thereby providing consistent illumination across the gap between the door and the baseboard. This retractable conduit allows the light strips to remain at a consistent distance from one another, regardless of the door's position, ensuring continuous lighting without any gaps.

(7) A key advantage of the present invention is its ability to operate using a single power source. The retractable conduit houses the necessary wiring that connects the baseboard light strip with the door light strip, eliminating the need for separate power sources. By utilizing a single power source, the present invention simplifies the installation process, reduces energy consumption, and enhances overall system efficiency. This centralized power system also minimizes the risk of power supply failures that could otherwise disrupt the lighting.

(8) The present invention solves the problems associated with the prior art by offering a streamlined solution for maintaining seamless lighting across the door and baseboard junction. By integrating the light strips on both the baseboard and door and linking them via a retractable conduit, the present invention ensures continuous illumination.

(9) Additionally, the present invention's single power source configuration significantly reduces the complexity found in the prior art, where multiple power sources are often required to operate separate lighting elements. The present invention not only simplifies the electrical setup but also contributes to energy savings and ease of maintenance. This feature makes it an ideal solution for modern interiors that demand both functionality and aesthetic appeal.

(10) In summary, the present invention provides a lighting system that addresses the limitations of the prior art. It offers seamless and consistent illumination between the bottom edge of a door and the adjacent baseboard, utilizing a concealed hinge, a retractable conduit, and a single power source. This innovative system enhances both the aesthetic and functional aspects of interior lighting, making it a practical and efficient solution for modern interior environments

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a bottom perspective view of the present invention.

(2) FIG. 2 is an electronic diagram of the present invention.

(3) FIG. 3 is a perspective view of the present invention.

(4) FIG. 4 is a bottom plan view of the present invention in the open configuration.

(5) FIG. 5 is a detail bottom plan view of the present invention in the open configuration.

(6) FIG. 6 is a bottom detail view of the proximal edge of the door of the present invention in the closed configuration.

(7) FIG. 7 is a bottom detail view of the distal edge of the door of the present invention in the closed configuration.

(8) FIG. 8 is a perspective view of an alternate embodiment of the present invention comprising a profile member wherein a portion of the baseboard is cut away to expose the profile member.

(9) FIG. 9 is a bottom perspective view of the present invention showing an alternate embodiment.

DETAIL DESCRIPTIONS OF THE INVENTION

(10) All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.

(11) As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the disclosure and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified as being “preferred” is considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other

embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

(12) Accordingly, while embodiments are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure, and are made merely for the purposes of providing a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself.

(13) Additionally, it is important to note that each term used herein refers to that which an ordinary artisan would understand such term to mean based on the contextual use of such term herein. To the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the ordinary artisan should prevail.

(14) Furthermore, it is important to note that, as used herein, “a” and “an” each generally denotes “at least one,” but does not exclude a plurality unless the contextual use dictates otherwise. When used herein to join a list of items, “or” denotes “at least one of the items,” but does not exclude a plurality of items of the list. Finally, when used herein to join a list of items, “and” denotes “all of the items of the list.”

(15) The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While many embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the appended claims. The present disclosure contains headers. It should be understood that these headers are used as references and are not to be construed as limiting upon the subjected matter disclosed under the header.

(16) Other technical advantages may become readily apparent to one of ordinary skill in the art after review of the following figures and description. It should be understood at the outset that, although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described below.

(17) Unless otherwise indicated, the drawings are intended to be read together with the specification, and are to be considered a portion of the entire written description of this invention. As used in the following description, the terms “horizontal”, “vertical”, “left”, “right”, “up”, “down” and the like, as well as adjectival and adverbial derivatives thereof (e.g., “horizontally”, “rightwardly”, “upwardly”, “radially”, etc.), simply refer to the orientation of the illustrated structure as the particular drawing figure faces the reader. Similarly, the terms “inwardly”, “outwardly” and “radially” generally refer to the orientation of a surface relative to its axis of elongation, or axis of rotation, as appropriate.

(18) The present disclosure includes many aspects and features. Moreover, while many aspects and features relate to, and are described in the context of a lighting system **1**, embodiments of the present disclosure are not limited to use only in this context.

(19) As shown in FIG. 1, the present invention is a lighting system **1** comprising a door **100**, a baseboard **400**, and a plurality of lighted strips **103** (composed of **103a** and **103b**). In the context of the present invention, the door **100** is hingedly engaged to the baseboard **400**, by which the door **100** hinges along an axis of rotation **304** in relation to the baseboard **400**. In the context of the present invention, the baseboard **400** is defined as an adjacent member by which the door **100** hinges about. In the context of the invention, the baseboard **400**, as disclosed herein, may further comprise a cabinet comprising a hinged door; in such a context, the cabinet is considered the baseboard **400**. Furthermore, in the context of the present invention, the baseboard **400** may further be referred to herein as its ordinary definition being a member traversing along the lowest part of an interior wall, specifically positioned at the junction of the floor board and the wall. In the preferred embodiment of the present invention, a plurality of lighted strips **103** traverse each of: the door **100** and the baseboard **400**.

(20) In the preferred embodiment of the present invention, as shown in FIG. 1, the door **100** comprises a downwardly facing surface **101**, a channel **102**, a proximal edge **110**, and a distal edge **112**. Within the context of the present invention, the downwardly surface **101** of the door **100** is understood as being the portion of the door **100** that engages with a threshold of a doorway. In the preferred embodiment of the present invention, as shown in FIG. 1, the channel **102** is a recess traversing along the downwardly facing surface **101** of the door **100**. In the preferred embodiment of the present invention, the channel **102** traverses along the downwardly facing surface **101** of the door **100** from the distal end **112** to the proximal end **110**. In the context of the present invention, the distal end **112** and the proximal end **110** are opposing ends of the door **100**, known within the art as the portions of the door **100** that engage with the vertical portions of a door frame (the jamb). In the context of the present invention, the proximal edge **110** of the door **100** refers to the end that is hingedly connected to the baseboard **400**, wherein the distal end **112** is the opposing end, typically known within the art as the portion of the door **100** comprising a latch. Furthermore, within the preferred embodiment of the present invention, the proximal edge **110** of the door **100** comprises a cutout **111**. In the preferred embodiment of the present invention, the door **100** comprises two channels **102** traversing both sides of the door (front side and rear side). In the preferred embodiment of the present invention, the lighted strip **103** comprising a plurality of lights **104**, traverses the entirety of each channel **102**. In some embodiments of the present invention, the channel **102** further comprises a diffuser **105**, positioned beneath the lighted strip **104**, thereby diffusing the light emitted from the lighted strip **104**, creating a consistently “smooth” illumination pattern.

(21) As further shown in FIG. 1, in the preferred embodiment of the present invention, the baseboard **400** comprises an overhang **401** and an interior portion **410**. In the context of the present invention, the overhang **401** traverses horizontally about the baseboard **400** wherein an the overhang **401** composes a downwardly facing surface above the floor. In the preferred embodiment of the present invention, the overhang **401** comprises a channel **402** wherein the channel **402** is a recess traversing the entirety of the overhang **401**. In the preferred embodiment of the present invention, the lighted strip **103** is inset into the channel **402**, traversing the entirety of the channel **402**. In some embodiments of the present invention, the channel **402** of the overhang **401** also comprises a diffuser **105**. In the context of the present invention, the channel **402** of the overhang **401** horizontally traverses the entirety of the overhang **401** from the interior portion **410** of the baseboard **400**. In the context of the present invention, the interior portion **410** of the baseboard **400** is the portion of the baseboard **400** that is hingedly coupled to the proximal edge **110** of the door **100**. In the preferred embodiment of the present invention, the interior portion **410** of the baseboard **400** comprises a cutout **411** wherein the cutout **111** of the door **100** and the cutout **411** of the baseboard **400** are complimentary.

(22) Additionally, as also shown in FIG. 1, in the preferred embodiment of the present invention, the lighting system **1** further comprises a conduit **200** and a hinge **300**. In the context of the present

invention, the conduit **200** is a freely retractable sheath that communicates electricity between the plurality of lighted strips **103**. Furthermore, the hinge **300**, in the preferred embodiment, is a hidden hinge comprising a hinge member **302** and a cavity **303**. In the context of the present invention, the hinge member **302** is coupled to each of the door **100** and the baseboard **400**, whereby the cavity **303** is a recess cut into both the door **100** and the baseboard **400**, wherein the hinge member **302** moves freely within the cavity **303** enabling hinged motion about the axis of rotation **304**. As shown in FIG. 2, the conduit **200** communicates electricity, generated from a power source **500**, between the door **100** and the baseboard **400**. In the preferred embodiment of the present invention the power source is a 12V converter box.

(23) In the preferred embodiment of the present invention, as shown in FIG. 3, the hinge **300** and the conduit **200** are positioned on the proximal end **110** of the door **100** and the interior portion **410** of the baseboard **400**, thereby engaging said proximal end **110** and interior portion **410** with each other upon hinged motion. In the preferred embodiment, as shown in FIG. 4, FIG. 5, FIG. 6, and FIG. 7, the present invention comprises an open configuration **11** and a closed configuration **12**. In the context of the present invention, the open configuration **11**, as understood by those in the art, is defined as a position when the door is opened; wherein the closed configuration **12** is defined as the position when the door is fully closed. In the context of the present invention, when in the open configuration **11**, as shown in FIG. 4 and FIG. 5, the conduit **200** and the hinge **300** are exposed in between the proximal edge **110** of the door **100** and the interior portion **410** of the baseboard **400**.

(24) As shown in FIG. 6 and FIG. 7, in the closed configuration **12**, the hinge mechanism **302** withdraws into the cavity **303**; and the conduit **200** withdraws into the door **100** and the baseboard **400**. Furthermore, in the closed configuration **12**, the lighted strips **103** traversing the channel **102** of the door **100** and the lighted strips **130** traversing the channel **402** of the overhang **401** are directly, also referred to as “immediately,” adjacent to each other. In the context of the present invention, “directly adjacent” and “immediately adjacent” are defined as a distance between adjacent lights **106** between the plurality of lights is less than or equal to the distance between adjacent lights **106** on one of the plurality of lighted strips, even between the door **100** and the baseboard **400**. For example, the distance between the closest lights **106**, adjacent to each other on the door **100** and the baseboard **400**, is less than or equal to the distance between two adjacent lights **106** on the lighted strip **103a** traversing the channel **102** of the door **100** itself. In the context of the present invention, the retractable functionality of the hinge **300** and the conduit **200** provide a mechanism for the plurality of lighted strips **103** to be immediately adjacent to each other.

Moreover, in the preferred embodiment of the present invention, the complimentary cutouts **111,411**, engage each other when in the closed configuration, thus proving a seemingly continuous lighted strip between the proximal end **110** of the door **100** and the interior portion **410** of the baseboard **400**. In the preferred embodiment of the present invention, the baseboard **400** further comprises a door frame **412** wherein the distal portion **112** of the door **100** engages with the door frame **412**. In the context of the present invention, the door frame **412** is also referred to as a jamb. Similarly to the plurality of lighted strips **103** between the proximal edge **110** of the door **100** and the interior portion **410** of the baseboard **400**, in the preferred embodiment of the present invention, while in the closed configuration **12**, the distal edge **112** of the door **100** is directly adjacent to the door frame **412** of the base board **400**, thus aligning the plurality of lighted strips **103** on the door **100** and the baseboard **400**, specifically the door frame **412**, colinearly whereby each of the plurality of lighted strips **103** are immediately adjacent to each other.

(25) In some embodiments of the present invention, as shown in FIG. 8, the lighting system **1** comprises a plurality of profile members **600** wherein one profile member **600** is coupled to the door **100** and another profile member **600** is coupled to the baseboard **400**. In the context of the present invention, the profile member **600** comprises a body and a channel **602** wherein the body is coupled to either the door **100** or the baseboard **400**, the channel **602** protruding outwardly from the body, horizontally traversing the body. In the preferred embodiment of the present invention, the

channel **602** comprises a lighted strip **103**, comprising a plurality of lights **104**, traversing the channel **602**. In the preferred embodiment of the present invention, the profile member **600** is embedded into the door **100** and the baseboard **400**, as shown in FIG. **8**. In some embodiments of the present invention, the channel **602** of the profile member **600** comprises a diffuser **105** beneath the lighted strip **103**.

(26) In some embodiments of the present invention, as shown in FIG. **9**, the conduit **200** is a spring-loaded conductive member **201**, protruding from one of either: the proximal end **110** of the door **100** and the interior portion **410** of the baseboard **400**. In such embodiments, the spring-loaded conductive member **201** engages a complimentary conductive plate of the complimentary components, thereby communicating electricity across said spring-loaded conductive member **201**. In the context of the present invention, while in the open configuration, the spring-loaded conductive member **201** will be separated from the complimentary conductive plate, thereby maintaining an open circuit, only closing said circuit upon closure of the door **100**. Furthermore, in some embodiments of the present invention, the hinge mechanism **300** may transfer electricity between the plurality of lighted strips **103**. In such embodiments wherein the hinge mechanism **300** communicates electricity between the plurality of lighted strips **103**, different hinge mechanism **300** may provide positive and negative nodes.

(27) Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention.

Claims

1. A lighting system comprising: a door; a baseboard; a plurality of lighted strips; wherein: the door comprises a distal edge and a channel; the baseboard comprises an overhang and a door frame; one lighted strip of the plurality of lighted strips traverses the channel of the door up to the distal edge of the door; one lighted strip of the plurality of lighted strips traverses the overhang up to the door frame; the door hinging about an axis relative to the baseboard, composing a closed configuration wherein the door is inset into the doorframe and colinear with an adjacent portion of the baseboard; while in the closed configuration, the distal edge of the door is directly adjacent to the door frame of the base board, thus aligning the plurality of lighted strips colinearly whereby each of the plurality of lighted strips are immediately adjacent to each other.
2. The lighting system as claimed in claim 1, wherein the channel traverses a downwardly facing surface of the door.
3. The lighting system as claimed in claim 1, wherein the overhang comprises a channel, congruent to the channel of the door; wherein the respective lighted strip traversing the overhang, traverses the channel of the overhang.
4. The lighting system as claimed in claim 1, wherein: the door further comprises a proximal edge; the proximal edge hinges about the door frame; the proximal edge of the door and the door frame each comprise a cutout wherein said cutouts are complementary cutouts that engage with each other when the door is in the closed configuration.
5. The lighting system as claimed in claim 4 further comprising a hinge wherein aid hinge is coupled to the door and the door frame.
6. The lighting system as claimed in claim 5, wherein the hinge is a hidden hinge comprising: a hinge mechanism; and a cavity; wherein: the hinge mechanism is interposedly coupled to the door and the door frame; the cavity is composed of a recess within the door frame and a recess within the door; the hinge mechanism being contained within the cavity; the hinge mechanism being completely concealed within the cavity when in the closed configuration.
7. The lighting system, as claimed in claim 6, further comprising a conduit, wherein: the conduit is connected between the lighted strip traversing the channel of the door and the lighted strip

traversing the overhang; the conduit is freely held between the proximal edge of the door and the door frame, whereby the conduit retracts into one of either: the door and the door frame; upon hinged motion; and the conduit communicating electricity between the plurality of lighted strips.

8. The lighting system as claimed in claim 1 wherein the plurality of lighted strips comprises a diffuser.

9. The lighting system, as claimed in claim 7 wherein the conduit communicates electricity from a power source.

10. A lighting system comprising: a door; a baseboard; a conduit; and a plurality of lighted strips; wherein: the door comprises a distal edge and a channel; the baseboard comprises an overhang and a door frame; one lighted strip of the plurality of lighted strips traverses the channel of the door up to the distal edge of the door; one lighted strip of the plurality of lighted strips traverses the overhang up to the door frame; the conduit freely retractable within the door and the baseboard wherein the conduit connects to the lighted strip traversing the channel and to the lighted strip traversing the overhang; the door hinging about an axis relative to the baseboard, composing a closed configuration wherein the door is inset into the doorframe and colinear with an adjacent portion of the baseboard; while in the closed configuration, the distal edge of the door is directly adjacent to the door frame of the base board, thus aligning the plurality of lighted strips colinearly whereby each of the plurality of lighted strips are immediately adjacent to each other.

11. The lighting system as claimed in claim 10, wherein the channel traverses a downwardly facing surface of the door.

12. The lighting system as claimed in claim 11, wherein the overhang comprises a channel, congruent to the channel of the door; wherein the respective lighted strip traversing the overhang, traverses the channel of the overhang.

13. The lighting system as claimed in claim 12 wherein the door further comprises a proximal edge, wherein: the proximal edge hinges about the door frame; and the proximal edge of the door comprises a cutout.

14. The lighting system as claimed in claim 12 wherein the door frame further comprises a cutout, wherein: the cutout of the door frame is complementary to the cutout of the door; and the cutout of the door and the cutout of the door frame engage with each other when the door is in the closed configuration.

15. The lighting system as claimed in claim 14 further comprising a hinge wherein said hinge is coupled to the door and the door frame.

16. The lighting system as claimed in claim 15, wherein the hinge is a hidden hinge comprising: a hinge mechanism; and a cavity; wherein: the hinge mechanism is interposedly coupled to the door and the door frame; the cavity is composed of a recess within the door frame and a recess within the door; the hinge mechanism being contained within the cavity; the hinge mechanism being completely concealed within the cavity when in the closed configuration.

17. The lighting system as claimed in claim 1 wherein the plurality of lighted strips comprises a diffuser.

18. A lighting system comprising: a door; a baseboard; and a plurality of profile members; wherein: each profile member of the plurality of profile members comprises: a channel, wherein said channel comprises a lighted strip; the lighted strip traverses the channel one of the profile members is coupled to a bottom-most portion of the door; another profile member is coupled to a bottom-most portion of the baseboard; the door hinging about an axis relative to the baseboard, composing a closed configuration wherein the door is colinear with an adjacent portion of the baseboard; and in the closed configuration, the plurality of profile members are colinearly adjacent to each other.

19. The lighting system, as claimed in claim 18, further comprising a conduit, wherein: the conduit is connected between each lighted strip; the conduit is freely retractable within the door and baseboard; and the conduit communicating electricity between the plurality of lighted strips.

20. The lighting system as claimed in claim 1 wherein the plurality of lighted strips comprises a diffuser.
